NYU SHANGHAI · ENGLISH FOR ACADEMIC PURPOSES

Reading Packet

EAP 101: Systems Thinking

Spring 2027 · Katherine Tosi



Wk 1 What Is Systems Thinking?
Wk 2 Understanding Mental Models
Wk 3 The Iceberg Model
Wk 5 Time Delays in Systems
Wk 6 Limits to Growth vs. Success to the Successful
Wk 7 The Tragedy of the Commons
Wk 10 Leverage Points

Bring this packet to every class. Annotate it, argue with it, and write in the margins.

START HERE

How to use this packet

This course starts with you, then moves to your community and to global systems. These readings give you the vocabulary and tools for that journey. They are short on purpose: read slowly, more than once, and with a pen.

Before

Answer the **Before you read** question in one or two sentences. It switches on what you already know.

During

Underline key ideas. Write a question mark next to anything confusing and bring it to class. Check the key words at the end.

After

Complete the **task** and upload it to the **Student Folder** on the course Google Drive by **Sunday, 23:59**.

Readings are numbered by the week they belong to, so the numbers skip the weeks that have no reading. Each reading also lists that week's video and its pre-viewing question from the course website.

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WEEK 1 · READING 1

What Is Systems Thinking?

Adapted for EAP 101 from the work of Barry Richmond and Donella Meadows

Due Sun Jan 24, 23:59 | **About 90 min** | **Student Folder → Week 1**

BEFORE YOU READ

Think of a problem in your life that you have tried to fix more than once. Why did it keep coming back?

More than a pile of parts

Look around you. The building you are sitting in, the metro line that brought you here, the group chat buzzing on your phone, and the body that is reading this sentence are all systems. A **system** is a group of parts that are connected to each other and that together do something none of the parts could do alone.

Donella Meadows, one of the most influential systems thinkers, described every system as having three kinds of things: **elements** (the parts you can see and count), **interconnections** (the relationships that link the parts), and a **function or purpose** (what the whole system does) (Meadows, 2008). A pile of sand has elements, but almost no interconnections and no purpose: add or remove a handful and it is still a pile of sand. A basketball team is different. Change how the players pass to each other, or change what the team is trying to achieve, and you get a very different team, even with the same five people on the court.

That last point is the surprising one. In many systems, the elements are the least important part. A university replaces almost all of its students every four years, yet it keeps its own character. What stays the same is the structure: the rules, the relationships, the flows of information, and the shared purpose.

A definition to work with

Barry Richmond, a systems educator who did much to bring systems thinking into classrooms, offered this definition:

> *"Systems thinking is the art and science of making reliable inferences about behavior by developing an increasingly deep understanding of underlying structure."*
>
> **Richmond (1994)**

Three words in this definition need unpacking.

- **Behavior** means how something changes over time: a rising rent, falling grades, a friendship that slowly cools.
- **Structure** means how the parts are connected: who depends on whom, what causes what, and which rules shape people's choices.

- **Inference** means a conclusion you reach from evidence. Richmond asks for *reliable* inferences: conclusions you can test and trust, not guesses.

Notice also the phrase "increasingly deep." Systems thinking is never finished. Each time you look at a system, you should understand it a little better than you did before.

Lists versus loops

Most of us are trained to explain problems with lists. Why are students stressed? Too much homework, not enough sleep, too much time on phones. Each item on the list is treated as a separate cause pushing in one direction. Richmond (1993) called this "laundry list" thinking.

A systems thinker asks how the items on the list affect each other. Too much homework leads to later nights. Later nights lead to tiredness. Tiredness makes studying slower, so the same homework takes longer, which leads to even later nights. The causes are not separate. They form a **loop**, and the loop keeps the problem going. Richmond called this "closed-loop thinking," and it is the main habit this course will help you build.

Why it matters for you

When a problem keeps coming back, it is usually not because people are lazy or foolish. It is because the structure of the system keeps producing the same behavior. If you only react to the latest event, you will be reacting forever. If you understand the structure, you may find a place where a small change makes a lasting difference.

Over the next fourteen weeks you will practice this way of thinking on three levels: yourself, your community, and global challenges. You will start with the system you know best: you.

Three questions to carry with you

1. What are the parts, and how are they connected?
2. How is this system behaving over time?
3. What structure could be producing that behavior?

KEY WORDS

element: one part of a system that you can see or count

interconnection: a relationship that links two parts

purpose / function: what the whole system does or achieves

behavior: how something changes over time

structure: the way the parts are connected and organized

inference: a conclusion reached from evidence

feedback loop: a chain of causes that circles back to where it began

THIS WEEK'S VIDEO · LINKS ON THE COURSE WEBSITE

How the Modern World Makes Us Mentally Ill · The School of Life
Before you watch: Think of one problem people usually blame on themselves, such as stress or loneliness. Could something in society be part of the cause?

YOUR TASK · READING 1

Write **200–250 words** defining a system in your own words, with **one example from your daily life**. Name the elements, the interconnections, and the purpose of your example.

Tip: Name the elements, the interconnections, and the purpose of your example.

Upload: Student Folder → Week 1 · Due Sun Jan 24, 23:59

WEEK 2 · READING 2

Understanding Mental Models

Written for EAP 101

Due Sun Jan 31, 23:59 | About 90 min | Student Folder → Week 2

BEFORE YOU READ

Think of a time you were completely sure about something and then discovered you were wrong. What made you so sure?

The map in your head

Nobody carries the whole world in their head. Instead, we carry simplified pictures of how things work: how a good student behaves, what a professor wants, why prices rise, what makes a person trustworthy. The systems pioneer Jay Forrester put it simply: “The image of the world around us, which we carry in our head, is just a model” (Forrester, 1971).

Peter Senge (1990) called these pictures **mental models**: deeply held assumptions, generalizations, and images that shape both how we understand the world and how we act in it.

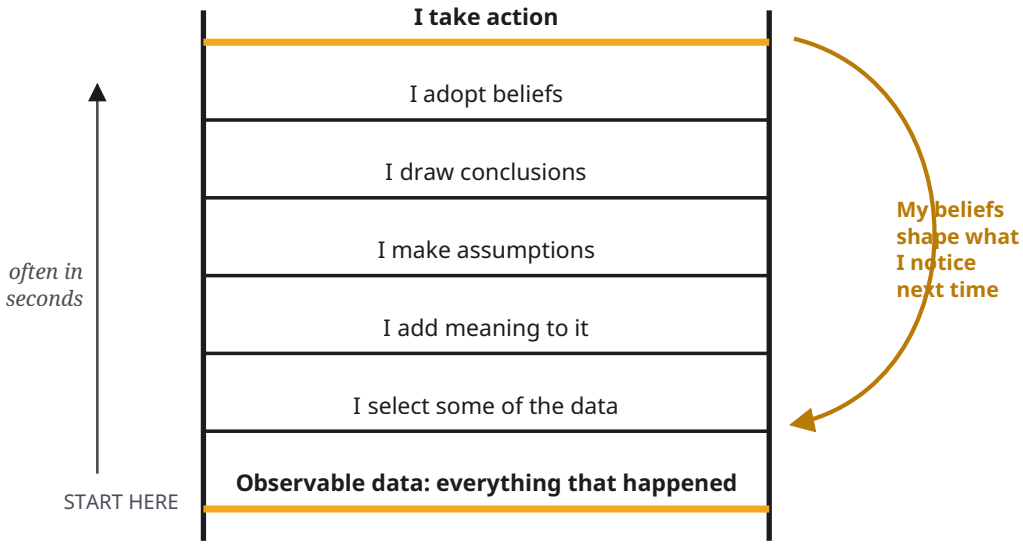
Mental models are not a weakness. They are necessary. Without them you would have to work out, from the beginning, how to cross a road or order a coffee. The problem is that mental models are always incomplete, and we rarely notice them. They feel like reality, not like a model of reality.

How mental models shape systems

Mental models matter for systems thinking because they drive decisions, and decisions build the structure of systems. Imagine a manager who believes that “people only work hard when they are watched.” She fills the workplace with monitoring. Employees feel distrusted, so they do only what is checked. The manager sees this and feels her belief has been proven right. The mental model has created the very behavior it predicted. This is called a **self-fulfilling** belief, and it is a loop.

Climbing the ladder of inference

The **ladder of inference**, developed by Chris Argyris and made popular in *The Fifth Discipline Fieldbook* (Senge et al., 1994), shows how quickly we move from what we observe to what we believe. Read it from the bottom up.



Here is how the climb can work. In the first week of a seminar, a classmate looks at her phone while you are speaking. That is the data you select. You decide she is bored by your idea (meaning), assume she thinks your English is weak (assumption), conclude that she does not respect you (conclusion), and come to believe that "people here don't take me seriously" (belief). In the next seminar you stay quiet (action), and now you notice every small sign of disinterest. In fact, she was answering a message from her mother. Almost nothing you believed came from the data. It came from the climb.

Mental models in an international classroom

Students arrive at university with different models of what a good student does. In some classrooms, a good student listens carefully and speaks only when sure of the answer. In others, silence suggests you have not prepared, and a good student thinks aloud, even when unsure. Neither model is wrong. Both are incomplete. In a seminar, they can lead people to misread each other completely. Naming your model ("In my high school, we usually…") is often the first step to working well together.

Catching your own mental models

You cannot see your mental models directly, but you can catch them with questions:

- What would have to be true for me to believe this?
- Which data did I select, and what did I ignore?
- What else could explain what I saw?
- Who might see this differently, and why?

KEY WORDS

mental model: a simplified picture in your mind of how something works

assumption: something you believe is true without checking

generalization: a statement about a whole group based on some examples

ladder of inference: a model of how we move from data to beliefs and actions

self-fulfilling: causing itself to come true

surface (verb): to bring something hidden into view

THIS WEEK'S VIDEO · LINKS ON THE COURSE WEBSITE

How Our Past Influences Our Present · The School of Life
Before you watch: What is one belief about learning or success that you have held since childhood? Where did it come from?

YOUR TASK · READING 2

Make **two-column notes**: key ideas about mental models on the left, an example from your own thinking on the right. Use the template below or copy it into your notebook.

Upload: Student Folder → Week 2 · Due Sun Jan 31, 23:59

TWO-COLUMN NOTES

Key idea about mental models	Example from my own thinking

WEEK 3 · READING 3

The Iceberg Model

Written for EAP 101

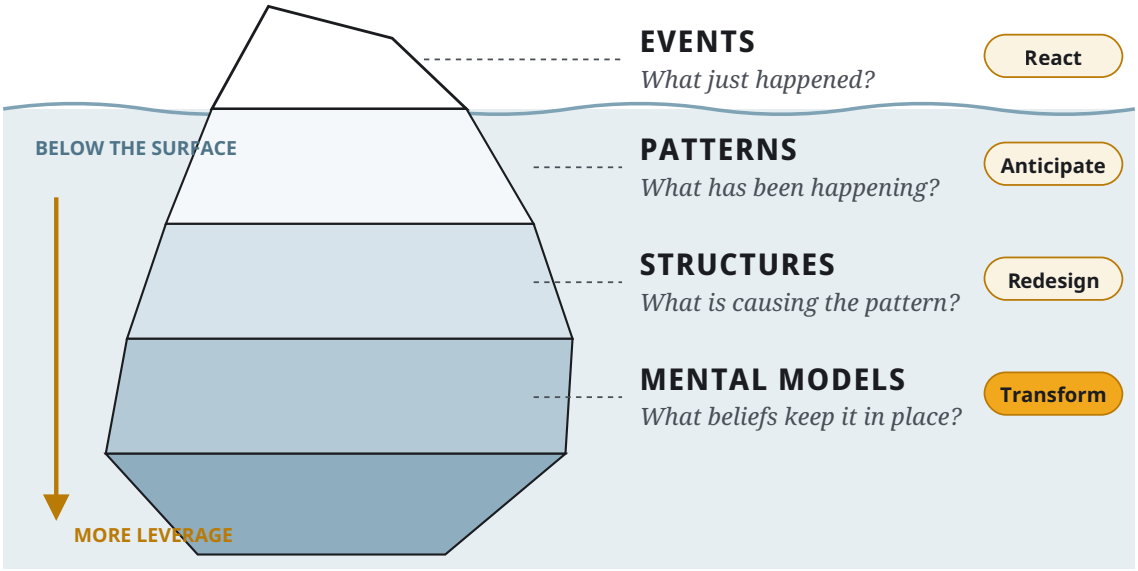
Due Sun Feb 14, 23:59 | About 75 min | Student Folder → Week 3

BEFORE YOU READ

Think of a news story you saw this week. What caused it? How far back can you trace the cause?

What you see is only the tip

Most of an iceberg is hidden under the water, and the hidden part is the part that sinks ships. Systems thinkers use the iceberg as a model for looking beneath the surface of any situation. The model has four levels, and each level answers a different question.



1 · Events

What we see and react to: a traffic jam, a failed exam, a flood, a viral post. Most news is made of events.

2 · Patterns and trends

When events repeat, they form patterns. Has this happened before? Is it getting better or worse? Patterns can be shown on a behavior-over-time graph.

3 · Structures

What produces the pattern. Structures include physical things (roads, buildings, apps), rules and policies (deadlines, prices, laws), and relationships and flows of information (who knows what, who depends on whom).

4 · Mental models

The beliefs, values, and assumptions that keep the structure in place. This is the deepest and least visible level.

The deeper you go, the more leverage you have

At the event level, the only option is to **react**: fix today's problem. At the pattern level, you can **anticipate** and prepare. At the structure level, you can **redesign** the system. At the mental-model level, you can **transform** it by changing the thinking that built the structure in the first place. Deeper changes are harder to make, but they last longer.

A worked example: late to class

Event	On Monday morning, six students arrive late to class.
Pattern	Lateness is worst on Mondays and on rainy days, and it has been the same all semester.
Structure	Many courses set weekly deadlines late on Sunday night, so students sleep late. The elevators are crowded just before class starts. On rainy days, more students take taxis and arrive at the same time.
Mental models	"A few minutes late doesn't matter." "Deadlines are more important than sleep." "Finishing at 23:58 counts as finishing on time."

Notice that a quick fix at the event level, such as locking the door at the start of class, may not change the pattern at all, because it does not touch the structure or the beliefs underneath it.

Using the iceberg on the news

1. **Choose one news event.** Write one sentence saying what happened.
2. **Look for the pattern.** Search for earlier, similar events. Is this new, or part of a trend?
3. **Ask about structure.** Which rules, incentives, physical conditions, or relationships make this pattern likely?
4. **Ask about mental models.** What do the people involved believe that keeps the structure in place?

Language for going deeper

Below the surface, you are making inferences, so use careful language:

- "This appears to be part of a larger pattern of…"
- "One possible structure behind this is…"
- "This may reflect a belief that…"
- "If this is true, we would expect to see…"

KEY WORDS

event: something that happens at one moment

pattern / trend: a series of events that repeats or changes in one direction

structure: what produces a pattern: physical things, rules, relationships

incentive: something that encourages a person to act in a certain way

leverage: the power to cause a big effect with a small action

transform: to change something completely and deeply

THIS WEEK'S VIDEO · LINKS ON THE COURSE WEBSITE

What It's Like Inside Our Minds · The School of Life
Before you watch: How much of what you think and feel are you aware of: most of it, about half, or very little? Why?

YOUR TASK · READING 3

Label **one news event** at all four levels: events, patterns, structures, mental models.

Tip: Include a link to your news source, and use the language box above for the deeper levels.

Upload: Student Folder → Week 3 · Due Sun Feb 14, 23:59

WEEK 5 · READING 5

Time Delays in Systems

Written for EAP 101 · includes stocks and flows

Due Sun Feb 28, 23:59 | About 75 min | Student Folder → Week 5

BEFORE YOU READ

Have you ever started something good for you, such as exercise or a new study method, and felt worse at first? How long did it take to get better?

The shower problem

You step into an unfamiliar hotel shower. The water is cold, so you turn the tap toward hot. Nothing happens, so you turn it further. Suddenly the water is burning hot. You turn it hard back toward cold. Nothing happens, so you turn it further, and soon you are freezing again.

The problem is not the shower, and it is not you. It is the **delay** between your action (turning the tap) and its result (the water changing temperature). When we do not account for a delay, we overreact, and the system swings back and forth, or **oscillates**. The lesson is simple and surprisingly hard to follow: make a small change, then wait.

Where delays hide

Delays appear at three points in almost every system:

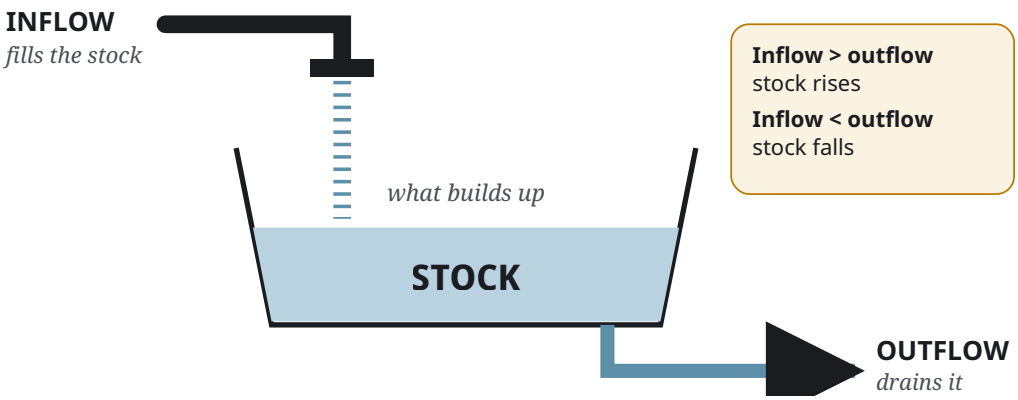
- **Between a problem and noticing it.** A health problem can grow for years before there are symptoms.
- **Between noticing and deciding.** Committees meet, budgets are approved, people argue.
- **Between deciding and seeing results.** A new metro line takes years to build. Study you do tonight shows up in an exam weeks later.

Worse before better, better before worse

Delays explain why good decisions can look like bad ones at first, and why bad decisions can look good. Many **investments** get worse before they get better: when you change your writing process, your first essays may be slower and messier, because the benefit comes later. Many **quick fixes** work the other way: they get better before they get worse. An energy drink at midnight helps now; the cost arrives the next afternoon. Cramming helps you pass a quiz this week, then most of it is forgotten before the final.

Stocks and flows: what builds up

A second idea explains many delays. A **stock** is anything that builds up or drains away over time: money in a bank account, water in a bathtub, knowledge, trust, stress, or carbon dioxide in the atmosphere. A **flow** is the rate at which a stock changes. **Inflows** fill it; **outflows** drain it (Meadows, 2008).



Here is the rule that surprises most people: **if the inflow is bigger than the outflow, the stock keeps rising, even when the inflow is falling.** Turn the tap down a little and the bathtub still fills, only more slowly. Carbon dioxide works the same way. Even if global emissions begin to fall, the amount of carbon dioxide in the atmosphere keeps rising for as long as emissions remain greater than what oceans, forests, and soils remove.

Stocks change slowly, so they create delays and act as buffers. Trust between two people usually builds slowly, through many small actions, but it can drain quickly. Sleep debt works the same way: one early night does not refill a week of short sleep.

What systems thinkers do with delays

- Ask *when*, not only *what*: when will we see the effect of this action?
- Be patient with investments, and suspicious of instant results.
- Make smaller changes, then wait and observe before changing again.
- Draw a behavior-over-time graph to see the delay instead of guessing.

KEY WORDS

delay: time between a cause and its effect

oscillate: to swing back and forth between two states

quick fix: a solution that helps now but may cause problems later

stock: anything that builds up or drains over time

flow (inflow / outflow): the rate at which a stock fills or drains

buffer: a stock that protects a system from sudden change

accumulate: to build up gradually

THIS WEEK'S VIDEO · LINKS ON THE COURSE WEBSITE

Stocks and Flows · Climate Interactive
Before you watch: Your phone battery is a stock. What fills it, what drains it, and why does it never change instantly?

YOUR TASK · READING 5

Answer the question set. Give **one example where a delay made a problem worse before it got better**.

1. In your own words, explain why the shower problem happens. Use the words *delay* and *oscillate*.
2. Give one example of "better before worse" and one of "worse before better" from your own life.
3. Name one stock in your life. What are its inflows and outflows?
4. Use the bathtub to explain why a stock can keep rising even when its inflow is falling.
5. Describe one situation where a delay made a problem worse before it got better. How long was the delay, and what would you do differently now?

Upload: Student Folder → Week 5 · Due Sun Feb 28, 23:59

WEEK 6 · READING 6

Limits to Growth vs. Success to the Successful

Two systems archetypes · written for EAP 101

Due Sun Mar 7, 23:59 | About 90 min | Student Folder → Week 6

BEFORE YOU READ

Think of a café, app, or club that became very popular very fast. What happened next?

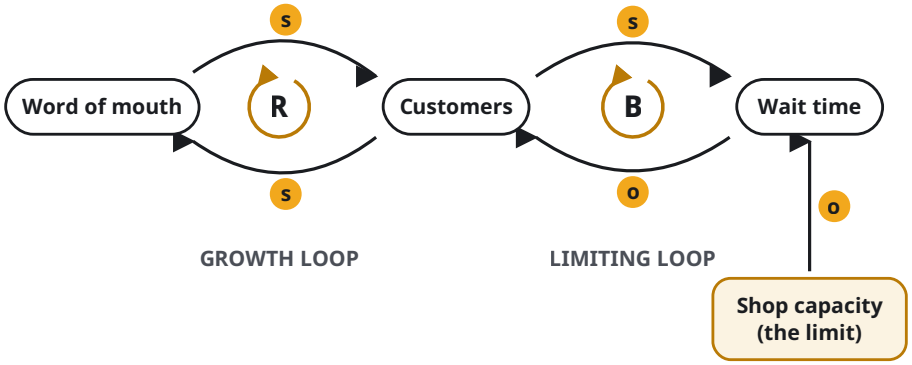
Patterns that repeat

Systems that look very different on the surface often share the same structure underneath. Peter Senge (1990) described a set of these common structures, called **systems archetypes**. Learn an archetype once and you will begin to see it everywhere: in friendships, companies, cities, and ecosystems. This week you will study two of them.

In the diagrams below, an **s** on an arrow means the two things move in the **same** direction (more leads to more, less leads to less). An **o** means they move in the **opposite** direction (more leads to less). **R** marks a reinforcing loop, which drives growth or decline. **B** marks a balancing loop, which pushes toward a limit or goal.

Archetype 1: Limits to Growth

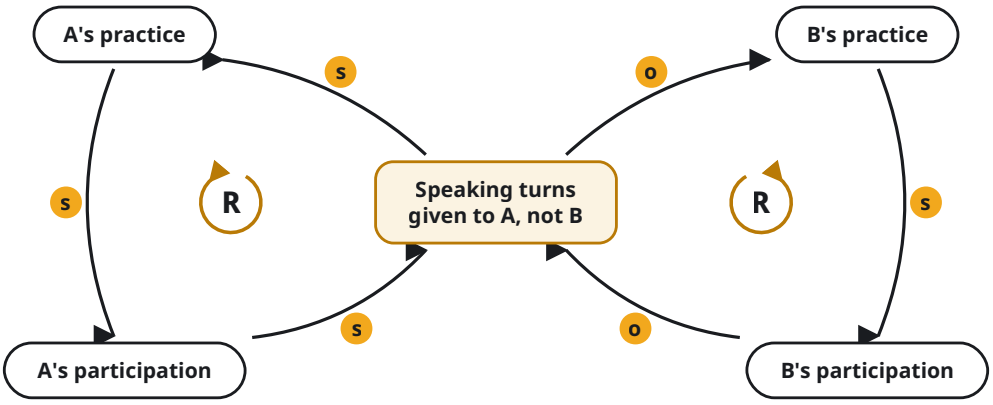
A new bubble tea shop opens near campus. The drinks are good, students post photos, their friends come, and more photos are posted. This is a **reinforcing loop**: success creates more success. But the shop can only make so many drinks an hour. As it becomes more popular, the queue grows. Long waits push customers away. This is a **balancing loop**, and it gets stronger as the shop approaches its limit. Growth slows, stops, and may even reverse.



The common mistake is to push harder on growth, for example with more advertising, which only makes the queue longer. **The leverage** is to find the limit early and work on it (more staff, online ordering), or to decide what size is sustainable. You can see the same archetype in a study method that works until you burn out, a city that grows until housing and traffic hold it back, or a group chat that is useful until there are too many messages to read.

Archetype 2: Success to the Successful

Two students, A and B, join a seminar with similar ability. In the first week, A happens to answer first. The response is warm, so A feels confident and speaks again, getting more practice and more feedback. B speaks less, gets less practice, feels less confident, and speaks even less. By Week 8, the gap looks like a difference in talent. In fact, the system created it.



In this archetype, two people or groups compete for a **limited resource**: attention, money, time, or space. Whoever has a little more success receives more of the resource, which makes them more successful still. Two reinforcing loops are linked: one rising, one falling. **The common mistake** is to assume the winner deserved it and the loser lacked ability. **The leverage** is to allocate resources according to shared goals rather than past success, or to give each side its own resources. In a seminar, that might mean structured turn-taking. The same structure appears in review apps that show popular restaurants first, in research funding, and in social media follower counts.

	Limits to Growth	Success to the Successful
Loops	One reinforcing loop meets one balancing loop	Two linked reinforcing loops
Over time	Fast growth, then slowing or stopping	A gap that keeps getting wider
Key question	What limit will this growth run into?	How are resources being shared, and why?
Leverage	Work on the limit, not the growth	Allocate by goals, not past success

KEY WORDS

archetype: a common pattern or structure that appears in many systems

reinforcing loop (R): a loop that pushes change further in the same direction

balancing loop (B): a loop that slows change or pushes toward a limit

allocate: to give out resources for a particular purpose

sustainable: able to continue over a long period

constraint: something that limits what is possible

THIS WEEK'S VIDEO · LINKS ON THE COURSE WEBSITE

Limits to Growth Systems Archetype · Archetype explainer
Before you watch: Think of something that grew quickly and then stopped growing. What slowed it down?

Success to the Successful Systems Archetype · Archetype explainer
Before you watch: Why do people or teams who are already winning often keep winning?

YOUR TASK · READING 6

Compare the two archetypes in a table. Add **one real example of each** from campus, a city, or a company.

Upload: Student Folder → Week 6 · Due Sun Mar 7, 23:59

COMPARISON TABLE

	Limits to Growth	Success to the Successful
How it works		
What it looks like over time		
Real example (campus, city, or company)		
Where the leverage is		

WEEK 7 · READING 7

The Tragedy of the Commons

Written for EAP 101 · drawing on Hardin (1968) and Ostrom (1990)

Due Sun Mar 14, 23:59 | About 75 min | Student Folder → Week 7

BEFORE YOU READ

Think of a shared kitchen, fridge, or study room. Why does it sometimes become messy or overcrowded?

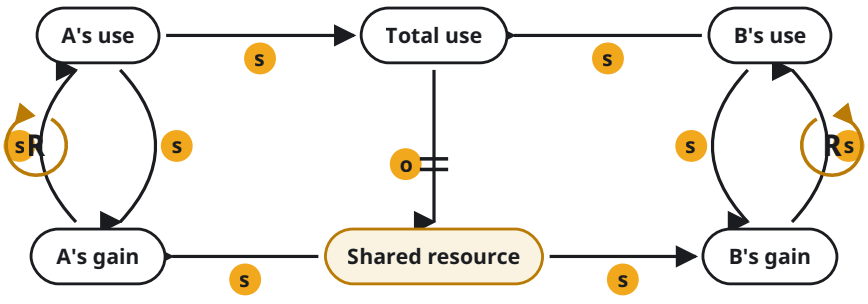
Hardin's pasture

In 1968, the ecologist Garrett Hardin published an essay that gave this archetype its name. He asked readers to imagine a pasture open to all herders. Each herder asks: should I add one more animal? If I do, I receive the full benefit of that animal. The cost, a little more overgrazing, is shared by everyone. So for each individual, adding an animal seems rational. But every herder reaches the same conclusion, the pasture is destroyed, and everyone loses. As Hardin (1968) put it, “Freedom in a commons brings ruin to all” (p. 1244).

The structure of a commons

A **commons** is a shared resource that many people can use, such as a fishery, a river, the atmosphere, or the quiet in a library. Its systems structure has five parts:

- **A shared resource**, which is a stock that can be used up.
- **Several users**, each with their own reinforcing loop: the more they use, the more they gain, so the more they use.
- **Private gain, shared cost.** Each user feels their own gain clearly, but the cost is spread across everyone.
- **A delay.** Damage to the resource appears slowly, so by the time it is visible, it is hard to reverse.
- **A final balancing effect.** When the resource collapses, everyone's gain falls at once.



The double line (//) marks a delay: damage to the resource appears slowly.

Commons are everywhere

Fish stocks, groundwater, and the atmosphere are global commons. Closer to home, so are dorm Wi-Fi in the evening, the shared fridge, the printer before a deadline, and the effort in a group project, where one member can do less and still share the grade.

Is the tragedy inevitable?

Hardin believed there were only two solutions: private ownership or control by government. The political economist Elinor Ostrom disagreed. She studied real communities, including fishing villages, irrigation systems, forests, and mountain pastures, that had managed shared resources successfully for long periods without either solution. Her research (Ostrom, 1990) earned her the 2009 Nobel Memorial Prize in Economic Sciences; she was the first woman to receive it. Her design principles, simplified, include:

- **Clear boundaries:** everyone knows who may use the resource.
- **Rules that fit local conditions**, not one rule for everywhere.
- **Users help make the rules**, so they understand and accept them.
- **Monitoring** by people who are accountable to the users.
- **Graduated sanctions:** small penalties first, larger ones for repeated misuse.
- **Fast, fair ways to resolve conflicts.**

In systems terms, each principle adds a **balancing loop**: feedback that makes overuse visible and costly early, before the delay hides the damage.

Language for proposing a change

- "One way to protect this commons would be to…"
- "This could work because it adds feedback that…"
- "A possible objection is…; however,…"

KEY WORDS

commons: a shared resource that many people can use

rational: based on clear reasoning, from one person's point of view

overgrazing: when animals eat plants faster than they can grow back

free rider: someone who benefits without contributing

monitoring: regularly checking what is happening

sanction: a penalty for breaking a rule

THIS WEEK'S VIDEO · LINKS ON THE COURSE WEBSITE

What is the tragedy of the commons? · TED-Ed · Nicholas Amendolare
Before you watch: Name one resource everyone on campus shares. What happens if a few people take too much?

YOUR TASK · READING 7

Identify the **shared resource**, the **users**, and the **feedback loop**. Suggest **one change** that could protect the commons.

Upload: Student Folder → Week 7 · Due Sun Mar 14, 23:59

WEEK 10 · READING 10

Leverage Points

A reading guide to Donella Meadows, "Leverage Points: Places to Intervene in a System" (1999)

Due Sun Apr 11, 23:59 | **About 90 min** | **Student Folder → Week 10**

BEFORE YOU READ

Think of a rule at school or at home that, if it changed, would change many other things. Why does that rule have so much power?

About this reading

Donella Meadows (1941–2001) was an environmental scientist, a lead author of *The Limits to Growth* (1972), and a teacher. Her essay "Leverage Points: Places to Intervene in a System" is one of the most widely read texts in systems thinking. She describes leverage points as places within a complex system where "a small shift in one thing can produce big changes in everything" (Meadows, 1999).

Read the original essay online: donellameadows.org/archives/leverage-points-places-to-intervene-in-a-system/
Selected sections: the introduction; points 12, 6, 5, 3, and 2; and the closing section. Read the other points if you have time.

The surprise in Meadows' list

Meadows ranked twelve places to intervene, from the weakest (12) to the strongest (1). Her key insight is that the places people usually push, such as budgets, prices, and targets, are often the weakest. The strongest leverage points, such as the goals of a system and the beliefs it grows from, are deeper, harder to see, and harder to change. She also noticed that people often find the right leverage point and then push it in the wrong direction.

The twelve points in plain English

#	Leverage point	Campus example
12	**Numbers:** amounts such as prices, taxes, budgets, and limits	*Raising the free printing allowance*
11	**Buffers:** the size of stocks that keep a system stable	*Extra library seats during exam weeks*
10	**Physical structure:** the layout of stocks and flows	*Where elevators and stairs are placed*
9	**Delays:** how long it takes for the system to respond	*How quickly feedback on essays is returned*
8	**Balancing loops:** the strength of feedback that corrects problems	*Course surveys that actually lead to changes*

7	**Reinforcing loops:** the strength of loops that drive growth	*Limiting how much the biggest clubs can keep growing*
6	**Information flows:** who can see what information	*Showing each dorm room its own electricity use*
5	**Rules:** incentives, punishments, and constraints	*Attendance and grading policies*
4	**Self-organization:** the power to change the system's own structure	*Students allowed to create new clubs and courses*
3	**Goals:** what the system is trying to achieve	*"Rank highly" versus "every student learns"*
2	**Paradigms:** the shared beliefs the system grows from	*"Education is a competition" versus "education is shared growth"*
1	**Transcending paradigms:** staying open to the idea that no paradigm is the whole truth	*Being able to question your own deepest assumptions*

Point names are simplified; examples are written for EAP 101. Check each point against Meadows' own explanation as you read.

Strategies for a challenging text

Meadows' essay is dense, with long sentences, technical terms, and many examples from economics and ecology. You do not need to understand every word.

- Read the table above first, so you know where the essay is going.
- For each point you read, write one sentence in your own words and one example.
- Mark confusing passages with "?" and bring them to class, instead of stopping.
- Notice her examples. They often explain the idea better than the definitions.

KEY WORDS

intervene: to act in order to change what is happening

leverage point: a place where a small change can cause large effects

parameter: a number that sets how a system runs, such as a price or limit

paradigm: a set of deep shared beliefs about how the world works

self-organization: a system's ability to change its own structure

transcend: to go beyond the limits of something

THIS WEEK'S VIDEO · LINKS ON THE COURSE WEBSITE

Leverage Points: Places to Intervene in a System · Community Climate Coaches
Before you watch: If you could change one thing about how your university works, which change would make the biggest difference?

YOUR TASK · READING 10

Read **Donella Meadows, *Leverage Points*** (selected sections). Choose **one leverage point for your project topic** and explain why it could work.

Upload: Student Folder → Week 10 · Due Sun Apr 11, 23:59

KEEP THIS PAGE OPEN

Glossary

Every key word in the packet, in alphabetical order. The tag shows the reading where it first appears.

accumulate R5
to build up gradually

allocate R6
to give out resources for a particular purpose

archetype R6
a common pattern or structure that appears in many systems

assumption R2
something you believe is true without checking

balancing loop (B) R6
a loop that slows change or pushes toward a limit

behavior R1
how something changes over time

buffer R5
a stock that protects a system from sudden change

commons R7
a shared resource that many people can use

constraint R6
something that limits what is possible

delay R5
time between a cause and its effect

element R1
one part of a system that you can see or count

event R3
something that happens at one moment

feedback loop R1
a chain of causes that circles back to where it began

flow (inflow / outflow) R5
the rate at which a stock fills or drains

free rider R7
someone who benefits without contributing

generalization R2
a statement about a whole group based on some examples

incentive R3
something that encourages a person to act in a certain way

inference R1
a conclusion reached from evidence

interconnection R1
a relationship that links two parts

intervene R10
to act in order to change what is happening

ladder of inference R2
a model of how we move from data to beliefs and actions

leverage R3
the power to cause a big effect with a small action

leverage point R10
a place where a small change can cause large effects

mental model R2
a simplified picture in your mind of how something works

monitoring R7
regularly checking what is happening

oscillate R5
to swing back and forth between two states

overgrazing R7
when animals eat plants faster than they can grow back

paradigm R10
a set of deep shared beliefs about how the world works

parameter R10
a number that sets how a system runs, such as a price or limit

pattern / trend R3
a series of events that repeats or changes in one direction

purpose / function R1
what the whole system does or achieves

quick fix R5
a solution that helps now but may cause problems later

rational R7
based on clear reasoning, from one person's point of view

reinforcing loop (R) R6
a loop that pushes change further in the same direction

sanction R7
a penalty for breaking a rule

self-fulfilling R2
causing itself to come true

self-organization R10
a system's ability to change its own structure

stock R5
anything that builds up or drains over time

structure R1
the way the parts are connected and organized

surface (verb) R2
to bring something hidden into view

sustainable R6
able to continue over a long period

transcend R10
to go beyond the limits of something

transform R3
to change something completely and deeply

SOURCES

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Readings in this packet are written or adapted for EAP 101. Short quotations are used for teaching and are credited above. Diagrams were drawn for this packet.